NATIONAL UNIVERSITY OF SINGAPORE
DEPARTMENT OF ELECTRICAL & COMPUTER ENGINEERING

ACADEMIC YEAR 2025-2026
SEMESTER 1

EE2213: Introduction to AI

ASSIGNMENT 1: L2-L3

Assignment 1 (4% of total grade)
Submission Deadline: 5 September 2025 (Friday), 23:59

Problem Statement:

Given two arbitrary vectors $x \in \mathcal{R}^m$ and $y \in \mathcal{R}^m$, write a Python routine to compute their Euclidean distance and Manhattan distance.

Your function should accept any two 1D NumPy arrays and return both distances as **NumPy arrays**.

Hint: You might find the NumPy library ('import numpy as np') and its operations helpful.

Instructions:

Submit a file named **A1_StudentMatriculationNumber.py** (replace "StudentMatriculationNumber" with your own your student matriculation number).

Your Python code as a function routine (def A1_StudentMatriculationNumber (x, y)) that takes x and y as inputs and returns:

- The Euclidean distance
- The Manhattan distance

You are **not allowed to use** np.linalg.norm, scipy.spatial.distance, or any other high-level distance functions. Please compute both Euclidean and Manhattan distances using only basic NumPy operations (e.g., element-wise subtraction, squaring, absolute value, sum, and square root). This restriction is intended to help you better understand how distance metrics work and to give you practice with vector and array manipulation in NumPy.

Your function should return both distances rounded to two decimal places.

Do not redefine x or y inside the function. We will do it in the main calling part during grading.

Your function must check that x and y are:

- 1D arrays, and
- of the same length

If either condition is not met, raise an informative error message.

Rename both your file and your function using your student matriculation number.
For example, if your matriculation ID is **A1234567R**, then:

- Filename should be **"A1_A1234567R.py"**
- Function name should be **"A1_A1234567R"**

Please use the provided template (A1_StudentMatriculationNumber.py) and do not comment out any lines.

Please do NOT zip/compress your file.

Please ensure you test your code before submission. You can do this by running the provided file, A1_test.py, to confirm your code produces the correct output for the included test cases. Note that for grading, we will use a different and more comprehensive set of test cases.

You must follow instructions strictly; marks will be deducted otherwise due to large class size.

Marks Allocation:

- Correct Euclidean distance output: 2%
- Correct Manhattan distance output: 2%

The way we would run your code might be something like this (main calling part):

>> import A1_A1234567R as grading

>> #code to define x and y --- you do not need to redefine them inside your function

>> euclidean_dist, manhattan_dist = grading.A1_A1234567R (x, y)

Please ensure your filenames and function names follow the specified format exactly.

Important Notes:

Please **replace** both "StudentMatriculationNumber" and "MatricNumber" in your **filename and function name** with your **actual matriculation number**.

Submission folder: Canvas→EE2213→Assignments→A1

You may submit your assignment multiple times. We will take the last version submitted before the deadline as your final submission.

Deadline:
23:59, 5 September 2025 (Friday)
No extensions will be granted.

Late submission policy:

- Up to 12 hours late: 80% of your grade
- 1 day late: 50% of your grade
- 2 days late: 30% of your grade
- More than 2 days late: No marks (0%)